



Statistics of Financial Markets

Seminar 11: Fractal markets hypothesis and long memory



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Acronyms used in this material

- ▣ ACF: Autocorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ DFA: Detrended Fluctuation Analysis
- ▣ EMH: Efficient Market Hypothesis
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ FMH: Fractal Market Hypothesis
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GPH: Geweke–Porter–Hudak (log-periodogram estimator)
- ▣ MA: Moving Average
- ▣ OLS: Ordinary Least Squares
- ▣ SE: Standard Error

Today's Question and Route

- **Question:** how can we measure the long memory of a market without seeing memory where there is none?
 - ▶ this seminar comes **before** Lecture 11: the section "What You Need for Today" gives every definition the tasks use
- Route
 - ▶ Part A: R/S step by step, the Hurst exponent from an R/S plot, fractional Gaussian noise and risk scaling, ARFIMA weights and autocorrelations, on paper
 - ▶ Part B: R/S, DFA, GPH and Lo's test with Monte Carlo bands on the S&P 500, BET, Banca Transilvania and Bitcoin; long memory in volatility; rolling exponents; each with an interpretation question
 - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class

Exercise Map

Task	Question	Type	Model
A1, A2	R/S of eight returns step by step; H from three points of an R/S plot	Solved, Proposed	A1
A3, A4	fractional Gaussian noise: autocorrelations, 10-day risk, VaR 1% (value at risk)	Solved, Proposed	A3
A5, A6	ARFIMA(0, d ,0): weights and autocorrelations against an AR(1)	Solved, Proposed	A5
B1, B2	R/S, DFA, GPH and Lo's test with Monte Carlo bands: S&P 500; BET, TLV	Solved, Proposed	B1
B3, B4	long memory in $ r_t $ and r_t^2 , the shuffle test: S&P 500; Bitcoin	Solved, Proposed	B3
B5, B6	rolling DFA exponents with a Monte Carlo band: S&P 500; DAX, TLV	Solved, Proposed	B5
C1, C2	does memory change in crises? what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data Used

Series	Source	Price used	Period
S&P 500, DAX, BET	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
Banca Transilvania (TLV)	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped (except Bitcoin); the last day is 18 September 2026
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, [Chapter 2](#))
- In the notebook: `returns('bet')`, `hurst_rs(x)`, `hurst_dfa(x)`, `gph(x)`, `lo_test(x)`, `mc_null(N)`; no account or key is needed

What You Need for Today (1/5): the Fractal Market Hypothesis and H

- **FMH** (fractal market hypothesis) [Peters \(1994\)](#): a market is stable when investors with many horizons (minutes to decades) trade with each other
 - ▶ in a crisis, all horizons shrink to the shortest one: no buyers, no liquidity
 - ▶ testable through scaling and memory across horizons
- **Self-similarity**: $X_{ct} \stackrel{d}{=} c^H X_t$; the h -day standard deviation is h^H times the daily one
 - ▶ **Hurst exponent** $H \in (0, 1)$: $H = 0.5$ independent increments (random walk, EMH); $H > 0.5$ persistence; $H < 0.5$ anti-persistence
 - ▶ H is not a probability; the fractal dimension of a price path is $D = 2 - H$

What You Need for Today (2/5): the R/S Statistic

- Block $x_1, \dots, x_n$, mean $\bar{x}$; cumulative deviations $Y_k = \sum_{j=1}^k (x_j - \bar{x})$
 - ▶ range $R_n = \max_k Y_k - \min_k Y_k$; standard deviation $S_n = \sqrt{\frac{1}{n} \sum_j (x_j - \bar{x})^2}$; $(R/S)_n = R_n/S_n$ [Hurst \(1951\)](#)
- $(R/S)_n \approx c n^H$: average R/S over non-overlapping blocks of size n , then regress $\log(R/S)_n$ on $\log n$; the slope is $\hat{H}$
 - ▶ three block sizes equally spaced on a log scale: the OLS slope equals the slope between the outer points
- Small samples: for i.i.d. data, the R/S slope is above 0.5 (about 0.544 for $N = 6714$) [Anis & Lloyd \(1976\)](#)

What You Need for Today (3/5): Long Memory, fGn and ARFIMA

- **Long memory:** $\rho(k) \sim C k^{2d-1}$, $0 < d < 0.5$; the autocorrelations decay hyperbolically and are not summable (Franke, Härdle & Hafner (2019), 14.1); short memory (AR): $\rho(k) = \phi^k$
- **Fractional Gaussian noise (fGn)**, the increments of fractional Brownian motion Mandelbrot & Van Ness (1968):
 - ▶ $\rho(k) = \frac{1}{2} \left(|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H} \right) \approx H(2H-1)k^{2H-2}$; $\rho(1) = 2^{2H-1} - 1$
- **ARFIMA(0,d,0)** Hosking (1981): $(1-L)^d X_t = \varepsilon_t$, $d = H - 0.5$
 - ▶ $(1-L)^d = \sum_j \pi_j L^j$: $\pi_0 = 1$, $\pi_j = \pi_{j-1}(j-1-d)/j$; MA weights $\psi_j = \psi_{j-1}(j-1+d)/j$
 - ▶ $\rho(1) = d/(1-d)$, $\rho(k) = \rho(k-1)(k-1+d)/(k-d)$

What You Need for Today (4/5): DFA, GPH, Lo's Test and Monte Carlo Bands

- **DFA** Peng et al. (1994): profile Y_k ; straight line fitted in each block of size n ; $F(n)$ = root mean square of the deviations; slope of $\log F$ on $\log n = \hat{H}$
- **GPH** Geweke & Porter-Hudak (1983): OLS of $\ln I(\lambda_j)$ (periodogram) on $-\ln(4 \sin^2(\lambda_j/2))$, $j \leq m = \lfloor T^{0.5} \rfloor$; slope $\hat{d}$, SE $\pi/\sqrt{24m}$
- **Lo's modified R/S** Lo (1991): $V = R/(\sqrt{N}\hat{\sigma}(q))$ with a Newey–West variance; H_0 : short memory; reject at 5% if $V \notin [0.809, 1.862]$
- **Monte Carlo band** Veron (2002): simulate many i.i.d. Normal series of the same length, apply the same estimator, take the 2.5% and 97.5% quantiles
 - ▶ reject $H = 0.5$ only if $\hat{H}$ lies outside the band

What You Need for Today (5/5): Spurious Memory, Volatility, Rolling Windows

- ▣ **Spurious long memory:** mean shifts [Diebold & Inoue \(2001\)](#) and persistent GARCH volatility ([Chapter 9](#)) produce slowly decaying sample ACFs
- ▣ **Shuffle test:** a random permutation keeps the distribution and destroys the order; if $\hat{H}$ falls to 0.5, the memory was in the order of the days
- ▣ **Volatility:** long memory of $|r_t|$ and r_t^2 [Ding, Granger & Engle \(1993\)](#) concerns the size of returns, not their direction
- ▣ **Rolling windows:** $\hat{H}$ on windows of 1000 days moved by 21, dated at the window end; with 5% tests, about 5% of windows fall outside the band by chance

A1: R/S Step by Step [Solved]

Task

- ▣ Eight daily returns (%):
0.5; -0.3; 0.8; -0.2; 0.6; -0.1; 0.4; -0.7.
- ▣ 1. Compute the mean, the deviations and the cumulative deviations $Y_1, \dots, Y_8$.
- ▣ 2. Compute R , S (divisor n) and R/S .
- ▣ 3. S&P 500 returns since 2000 give $(R/S)_{16} = 3.91$, $(R/S)_{64} = 8.38$, $(R/S)_{256} = 17.03$; compute $\hat{H}$.
- ▣ Report: R , S , R/S , $\hat{H}$ and one sentence.

Solution

- ▣ 1. $\bar{x} = 0.125$; Y_k : 0.375, -0.050, 0.625, 0.300, 0.775, 0.550, 0.825, 0.000
- ▣ 2. $R = 0.825 - (-0.050) = 0.875$; $S = \sqrt{1.915/8} = 0.489$;
 $R/S = 1.788$
- ▣ 3. $\log_{10}$: n : 1.204, 1.806, 2.408; R/S : 0.592, 0.923, 1.231;
 $\hat{H} = 0.639/1.204 = 0.531$
- ▣ Alternating signs keep the cumulative deviations small; $\hat{H}$ slightly above 0.5 is the small-sample bias of R/S , not memory (B1).

A2: Grouped Signs [Proposed]

Solution discussed in class

Task

- ▣ Eight returns (%): 0.6; 0.4; 0.5; 0.3; -0.4; -0.6; -0.2; -0.5. Model: A1.
- ▣ 1. Compute the mean and $Y_1, \dots, Y_8$.
- ▣ 2. Compute R , S and R/S , and compare with A1.
- ▣ 3. BET returns give $(R/S)_{16} = 4.23$, $(R/S)_{64} = 10.08$, $(R/S)_{256} = 21.61$; compute $\hat{H}$.
- ▣ Report: R/S , $\hat{H}$ and two sentences.

A3: Fractional Gaussian Noise and 10-Day Risk [Solved]

Task

- ▣ Daily returns follow an fGn with $H = 0.7$ and daily volatility $\sigma = 1.2\%$; $z_{0.99} = 2.326$.
- ▣ 1. Compute $\rho(1)$ and $\rho(10)$ exactly and $\rho(10)$ with the approximation $H(2H - 1)k^{2H-2}$.
- ▣ 2. Compute the scaling factor 10^H and compare it with $\sqrt{10}$.
- ▣ 3. Compute the one-day and the 10-day VaR 1% (Normal) with h^H and with $\sqrt{h}$.
- ▣ 4. Compute the fractal dimension of the price path.
- ▣ Report: six numbers and one sentence.

Solution

- ▣ 1. $\rho(1) = 2^{0.4} - 1 = 0.320$;
 $\rho(10) = \frac{1}{2}(11^{1.4} - 2 \cdot 10^{1.4} + 9^{1.4}) = 0.0704$; approximation
 $0.7 \cdot 0.4 \cdot 10^{-0.6} = 0.0703$
- ▣ 2. $10^{0.7} = 5.012$ against $\sqrt{10} = 3.162$
- ▣ 3. $2.326 \times 1.2 = 2.79\%$; 10 days: $2.79 \times 5.012 = 13.99\%$
 against 8.83% (+58.5%)
- ▣ 4. $D = 2 - H = 1.30$
- ▣ Persistent returns make multi-day losses add up: the $\sqrt{h}$ rule understates 10-day risk by more than a third.

A4: Anti-Persistence and a Small Deviation from 0.5 [Proposed]

Solution discussed in class

Task

- ▣ The setting of A3 with $H = 0.4$ and with $H = 0.55$. Model: A3.
- ▣ 1. Compute $\rho(1)$ and $\rho(10)$ for both values of H .
- ▣ 2. Compute the 10-day VaR 1% for both and its difference from the $\sqrt{h}$ rule, in %.
- ▣ 3. Compute D for both.
- ▣ Report: eight numbers and one sentence.

A5: ARFIMA(0, 0.3, 0) against an AR(1) [Solved]

Task

- $(1 - L)^{0.3}X_t = \varepsilon_t$, ε_t i.i.d. $N(0, 1)$.
- 1. Compute π_1 , π_2 and the MA weights ψ_1 , ψ_2 .
- 2. Compute $\rho(1)$, $\rho(2)$, $\rho(10)$ and $\rho(50)$.
- 3. For an AR(1) with $\phi = \rho(1)$, compute $\rho(10)$ and $\rho(50)$.
- 4. Give H and the first lag with $\rho(k) < 0.05$ for both models.
- Report: ten numbers and one sentence.

Solution

- 1. $\pi_1 = -0.3$, $\pi_2 = -0.3 \times 0.7/2 = -0.105$; $\psi_1 = 0.3$, $\psi_2 = 0.3 \times 1.3/2 = 0.195$
- 2. $\rho(1) = 0.3/0.7 = 0.429$; $\rho(2) = 0.429 \times 1.3/1.7 = 0.328$; $\rho(10) = 0.173$; $\rho(50) = 0.091$
- 3. $0.429^{10} = 0.0002$; $0.429^{50} \approx 0$
- 4. $H = 0.8$; below 0.05 after 4 lags (AR) and 222 lags (ARFIMA)
- Same $\rho(1)$, very different memory: the AR(1) forgets in days, the ARFIMA in months.

A6: Strong Memory and Anti-Persistence [Proposed]

Solution discussed in class

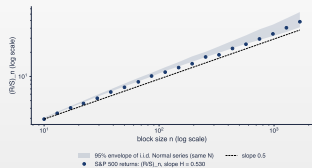
Task

- ▣ ARFIMA(0, d ,0) with $d = 0.45$ and with $d = -0.2$. Model: A5.
- ▣ 1. Compute π_1 , π_2 , $\rho(1)$, $\rho(2)$ and $\rho(10)$ for both.
- ▣ 2. For $d = 0.45$, compare $\rho(10)$ and $\rho(50)$ with an AR(1) with $\phi = \rho(1)$.
- ▣ 3. Give H for both and say which one is stationary.
- ▣ Report: a small table and one sentence.

B1: Long Memory in S&P 500 Returns? [Solved]

- ▣ **Question:** do daily S&P 500 returns have long memory?
- ▣ **Inputs:** S&P 500 closes since 2000, daily log returns in %
- ▣ **Tasks**
 1. Estimate H with R/S and DFA (block sizes from 10 to $N/4$) and d with GPH ($m = \lfloor T^{0.5} \rfloor$), with the standard error of $\hat{d}$.
 2. Compute Lo's modified statistic $V(q)$ with Andrews' q , and the classical statistic $V(0)$.
 3. Simulate 300 i.i.d. Normal series of the same length and give the 95% band of each estimator.
 4. Draw the R/S plot inside the Monte Carlo envelope.
 5. Interpretation: is the R/S estimate above 0.5 evidence of long memory?
- ▣ **Report:** a table of four estimates with their bands, the chart and two sentences
- ▣ **Notebook:** section B1

B1: Solution [Solved]



	R/S	DFA	GPH $\hat{H} = \hat{d} + 0.5$	Lo $V(q)$
estimate	0.530	0.488	0.536	1.559 ($q = 7$)
95% band	[0.512, 0.575]	[0.456, 0.540]	[0.363, 0.652]	[0.809, 1.862]

- $N = 6\,714$; GPH: $\hat{d} = 0.036$, SE 0.071 ($m = 81$), $t = 0.51$; classical $V(0) = 1.379$
- Interpretation: no: i.i.d. series of this length give R/S 0.544 on average; every estimate lies inside its band and Lo does not reject short memory

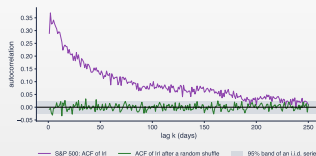
B2: BET and Banca Transilvania [Proposed]

- **Question:** does the BET or Banca Transilvania show long memory in returns? Model: B1.
- **Inputs:** BET since 2000, TLV since 2010 (without 30–31 May 2016); daily log returns in %
- **Tasks**
 1. Estimate H with R/S, DFA and GPH for each series.
 2. Compute Lo's $V(q)$ and say whether short memory is rejected at 5%.
 3. Compute the Monte Carlo band of each estimator for each length.
 4. Interpretation: what could explain the result for the BET?
- **Report:** one table and two sentences
- **Notebook:** section B2

B3: Long Memory in S&P 500 Volatility [Solved]

- ▣ **Question:** do the absolute and the squared S&P 500 returns have long memory?
- ▣ **Inputs:** S&P 500 daily log returns since 2000; $|r_t|$ and r_t^2
- ▣ **Tasks**
 1. Estimate d with GPH for $m = \lfloor T^{0.5} \rfloor$ and $m = \lfloor T^{0.6} \rfloor$, and H with DFA, for $|r_t|$ and r_t^2 .
 2. Report the ACF of $|r_t|$ at lags 1, 50 and 250 and the 95% band $\pm 1.96/\sqrt{T}$.
 3. Shuffle $|r_t|$ at random and repeat the estimates.
 4. Draw the ACF of $|r_t|$ before and after the shuffle.
 5. Interpretation: does long memory in $|r_t|$ contradict the weak form of the EMH?
- ▣ **Report:** a table, the chart and two sentences
- ▣ **Notebook:** section B3

B3: Solution [Solved]



	GPH $T^{0.5}$	GPH $T^{0.6}$	DFA	$\rho(1)$	$\rho(50)$	$\rho(250)$
$ r_t $	0.470	0.467	0.953	0.289	0.127	0.009
r_t^2	0.326	0.377	0.892	0.313	0.053	-0.010
$ r_t $ shuffled	0.028	-0.001	0.498	-0.008	-0.011	0.009

- $m = 81$ and 197 ; SE 0.071 and 0.046 ; ACF band ± 0.024 ; after the shuffle all estimates fall to 0 (d) or 0.5 (H)
- Interpretation: no: the weak form concerns the direction of returns (B1); long memory in $|r_t|$ means predictable risk, which needs a volatility model (Chapter 9)

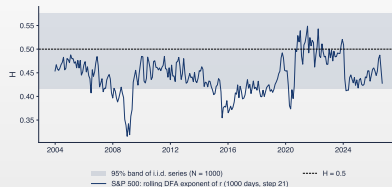
B4: Bitcoin Returns and Volatility [Proposed]

- **Question:** does Bitcoin show long memory in returns, in volatility, or in both? Models: B1, B3.
- **Inputs:** Bitcoin daily log returns since 2014 (7 days a week)
- **Tasks**
 1. Estimate R/S, DFA, GPH and Lo's $V(q)$ for the returns, with the Monte Carlo bands.
 2. Estimate GPH ($T^{0.5}$, $T^{0.6}$) and DFA for $|r_t|$ and r_t^2 .
 3. Repeat the volatility estimates after a random shuffle of $|r_t|$.
 4. Interpretation: is Bitcoin closer to the S&P 500 or to the BET?
- **Report:** one table and two sentences
- **Notebook:** section B4

B5: Rolling Hurst Exponents of the S&P 500 [Solved]

- ▣ **Question:** has the memory of S&P 500 returns changed since 2000?
- ▣ **Inputs:** S&P 500 daily log returns since 2000
- ▣ **Tasks**
 1. Estimate the DFA exponent on windows of 1000 days moved by 21 days, dated at the last day of each window.
 2. Compute the 95% Monte Carlo band for $N = 1000$.
 3. Report the minimum and the maximum with their dates, and the share of windows below and above the band.
 4. Draw the rolling exponent with the band.
 5. Interpretation: does a window outside the band prove that the market was inefficient at that time?
- ▣ **Report:** six numbers, the chart and two sentences
- ▣ **Notebook:** section B5

B5: Solution [Solved]



- ▣ 273 windows from 29 December 2003; band $[0.417, 0.577]$; mean $\hat{H} = 0.443$
- ▣ Minimum 0.316 (31 December 2008), maximum 0.549 (9 July 2021); 22.3% of the windows below the band, 0.0% above
- ▣ Interpretation: no: by chance about 14 of 273 windows fall outside, and consecutive windows overlap; only long runs count; here the runs are below the band (anti-persistence after 2008 and in 2016), never above

B6: Rolling Exponents of the DAX and of Banca Transilvania [Proposed]

- **Question:** does the result of B5 hold for the DAX and for Banca Transilvania? Model: B5.
- **Inputs:** DAX since 2000, TLV since 2010; daily log returns in %
- **Tasks**
 1. Estimate the rolling DFA exponent as in B5.
 2. Report the share of windows below and above the band, and the minimum and maximum with their dates.
 3. Compare the mean exponent in the first and in the last third of the windows.
 4. Interpretation: which market looks more stable over time?
- **Report:** one table and two sentences
- **Notebook:** section B6

C1: Does Market Memory Change in Crises? [Proposed]

- ▣ **Question:** does the Hurst exponent of returns or of volatility change in a crisis, as the fractal market hypothesis suggests?
- ▣ **Inputs:** S&P 500 and BET daily log returns; a calm year (2017) and two crisis years (September 2008–August 2009, February 2020–February 2021); models: B1, B3
- ▣ **Tasks**
 1. Estimate the DFA exponent of r_t and of $|r_t|$ and the annualised volatility in each of the three years, for both indices.
 2. Compute the Monte Carlo band for windows of about 250 days.
 3. List three reasons why one crisis year cannot settle the question.
 4. Interpretation: which design would separate a change of memory from a change of volatility?
- ▣ **Report:** a table and a plan for a project
- ▣ **Notebook:** section C1

C2: Audit an AI Answer [Proposed]

- ▣ A student asked an AI assistant to interpret long-memory estimates for the S&P 500 (daily, since 2000). The answer:
- ▣ (a) R/S gives $H = 0.530$ for the returns: above 0.5, so returns have long memory and trends can be exploited.
- ▣ (b) $H = 0.530$ means a 53% probability that tomorrow has the same sign as today.
- ▣ (c) GPH gives $d = 0.470$ for $|r|$, so $H = d - 0.5 = -0.030$: absolute returns are anti-persistent.
- ▣ (d) Long memory in $|r|$ contradicts the weak form of the EMH.
- ▣ (e) Lo's modified statistic $V = 1.56$ lies inside $[0.809, 1.862]$, so short memory is not rejected at 5%.
- ▣ (f) For more precision, use all $T/2 = 3357$ frequencies in the GPH regression.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

What You Should Take from Today

- ▣ R/S, DFA and GPH estimate the same H in different ways; report several, each with a Monte Carlo band for the same N
- ▣ $d = H - 0.5$; H is a scaling exponent, not a probability
- ▣ Returns of large markets: no long memory; volatility: long memory everywhere; the shuffle test locates it in the order of the days
- ▣ Rolling estimates: read long runs outside the band, not single windows
- ▣ An AI answer is a draft: check the band, the conversion between d and H , and the bandwidth

After the Seminar

- Lecture 11 develops each topic of today: the FMH, self-similarity, fBm and ARFIMA, the estimators, spurious memory, rolling exponents, risk scaling
- Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- C1 can grow into a team project: GARCH-standardised returns, many crises, a GARCH-based band
- Reading: [Franke, Härdle & Hafner \(2019\)](#), Ch. 14; [Lo \(1991\)](#); [Weron \(2002\)](#)

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